QP CODE: 25900334

MGU - UGP (HONOURS)REGULAR EXAMINATION, OCTOBER 2025

Third Semester

Discipline Specific Core Course(DSC) - MG3DSCCFS202 –COMPUTER SECURITY

(2024 ADMISSION ONWARDS)

Time:1.5Hours

Max marks: 50

Part A

Answer any **10** questions

Each question carries **1** marks

1. **Explain Clark – Wilson Model [U]/[CO1]**
   The Clark–Wilson model enforces data integrity through well-formed transactions and separation of duties using certification and enforcement rules.
2. **Define loss of availability. [K]/[CO1]**
   Loss of availability occurs when authorized users are unable to access information or system resources when needed.
3. **Define hashing [K]/[CO1]**
   Hashing is the process of transforming input data into a fixed-size hash value or digest using a hash function.
4. **Name the various Filtering Techniques used in firewall [K]/[CO1]**
   Packet filtering, Stateful inspection, Proxy filtering, and Application-level filtering
5. **Name the different types if Intrusion detection system. [U]/[CO1]**
   Host-based IDS (HIDS) and Network-based IDS (NIDS)
6. **Explain spyware [U]/[CO1]**
   Spyware is malicious software that secretly monitors user activity and collects information without consent
7. **Compare the role of symmetric encryption and asymmetric encryption [An]/[CO2]**
   Symmetric encryption uses a single shared key for both encryption and decryption.
   Asymmetric encryption uses a pair of keys — a public key for encryption and a private key for decryption.
8. **Examine how TLS protects against common security threats. [An]/[CO2]**

TLS protects against threats like eavesdropping, tampering, and message forgery by providing encryption, message integrity, and authentication between communicating parties.

9. **Explain one common vulnerability in web application. [U]/[CO1]**

   A common web application vulnerability is SQL Injection, where attackers insert malicious SQL code to manipulate or access a database illegally.

10. **Simplify the main difficulty faced in managing keys for database encryption. [An]/[CO2]**

    The main difficulty is secure key storage and distribution, ensuring that encryption keys are protected from unauthorized access while remaining accessible for authorized use.

11. **Summarize the main cause of SQLi attacks. [An]/[CO2]**

    Poor input validation in web applications

12. **Categorize tools for static analysis of mobile apps. [An]/[CO2]**

    Source code analysis tools – e.g., SonarQube, FindBugs, PMD
    Binary analysis tools – e.g., MobSF (Mobile Security Framework), AndroBugs, QARK . Give mark for any correct tool name

(10*1=10)

Part B

Answer any **4** questions
Each question carries **5** marks

13. **Explain the term access control policy [U]/[CO1]**

    Access control policy defines who (users or processes) can access what resources and under what conditions. It ensures confidentiality, integrity, and availability by restricting unauthorized access.Policies are based on principles like least privilege and need-to-know.Common types include Discretionary Access Control (DAC), Mandatory Access Control (MAC), and Role-Based Access Control (RBAC).
    Explanation, common types etc – 5 marks

14. **Distinguish between authentication and authorization. [An]/[CO2]**

    Authentication verifies the identity of a user or system (e.g., via password, biometrics).
    Authorization determines the permissions or access rights granted to an authenticated user.
    Definition ,purpose, example etc – 5 marks

15. **Discuss about different types of Phishing.** **[U]/[CO1]**

Email Phishing: Fake emails trick users into revealing personal or financial data.

Spear Phishing: Targeted phishing aimed at specific individuals or organizations.

Whaling: Phishing attacks directed at high-profile targets like executives.

Smishing: Phishing through SMS messages.

Vishing: Voice-based phishing attacks via phone calls pretending to be legitimate sources.

Explanation of different types -5 marks

16. **Write the uses of tunneling protocols in VPN.** **[A]/[CO3]**

Tunneling protocols encapsulate data packets to create a secure communication channel over public networks. Provide data confidentiality through encryption during transmission. Ensure data integrity and authentication of communicating parties. Common VPN tunneling protocols: PPTP, L2TP, IPsec, SSL/TLS, and OpenVPN. Used for remote access, secure communication, and bypassing geographic or network restrictions.

Explanation – 5 marks

17. **Explain how encryption protects mobile apps.** **[U]/[CO1]**

Encryption converts sensitive app data into unreadable form, protecting it from unauthorized access. Safeguards stored data (at rest) and transmitted data (in transit). Prevents data leakage if a device is lost or compromised. Protects communications between app and server using SSL/TLS encryption. Ensures compliance with security standards and user privacy protection.

Explanation – 5 marks

18. **Illustrate the dangers of week cryptography in web applications.**

[A]/[CO3]

Weak cryptography can be easily broken, leading to data breaches and unauthorized access. Enables attackers to decrypt sensitive information like passwords or financial data. Makes web applications vulnerable to man-in-the-middle (MITM) and replay attacks. Affects user trust and may lead to legal and reputational damage. Using strong, modern algorithms (e.g., AES, SHA-256) helps prevent such vulnerabilities

Explanaton – 5 marks

(4*5=20)

Part C

Answer any **two** questions

Each question carries **10** marks

19. **Discus s the various types of computer security threats.**
[U]/[CO1]

Malware, Phishing Attacks, Denial of Service (DoS) and Distributed DoS (DDoS), Man-in-the-Middle (MitM) Attacks, SQL Injection, Insider Threats, Spyware and Adware, Password Attacks, Zero-Day Exploits, Social Engineering

Explanation of any 7 – 10 marks

20. **Compare and contrast IPSec Authentication Header and Encapsulating Security Payload (ESP) in terms of confidentiality, authentication and integrity. [An]/[CO2]**

Authentication Header Provides authentication and integrity for IP packets.

ESP Provides confidentiality, authentication, and integrity.

Comparison – 5 marks

Difference – 5 marks

21. **Differentiate between cloud network access and traditional IT network access. [An]/[CO2]**

In Cloud Network Access, users access data, applications, and services hosted by a cloud provider over the Internet. In Traditional IT Network Access, resources are hosted locally on-premises within the organization's own infrastructure.

Any 5 Difference with explanation - 10 marks

(2*10=20)